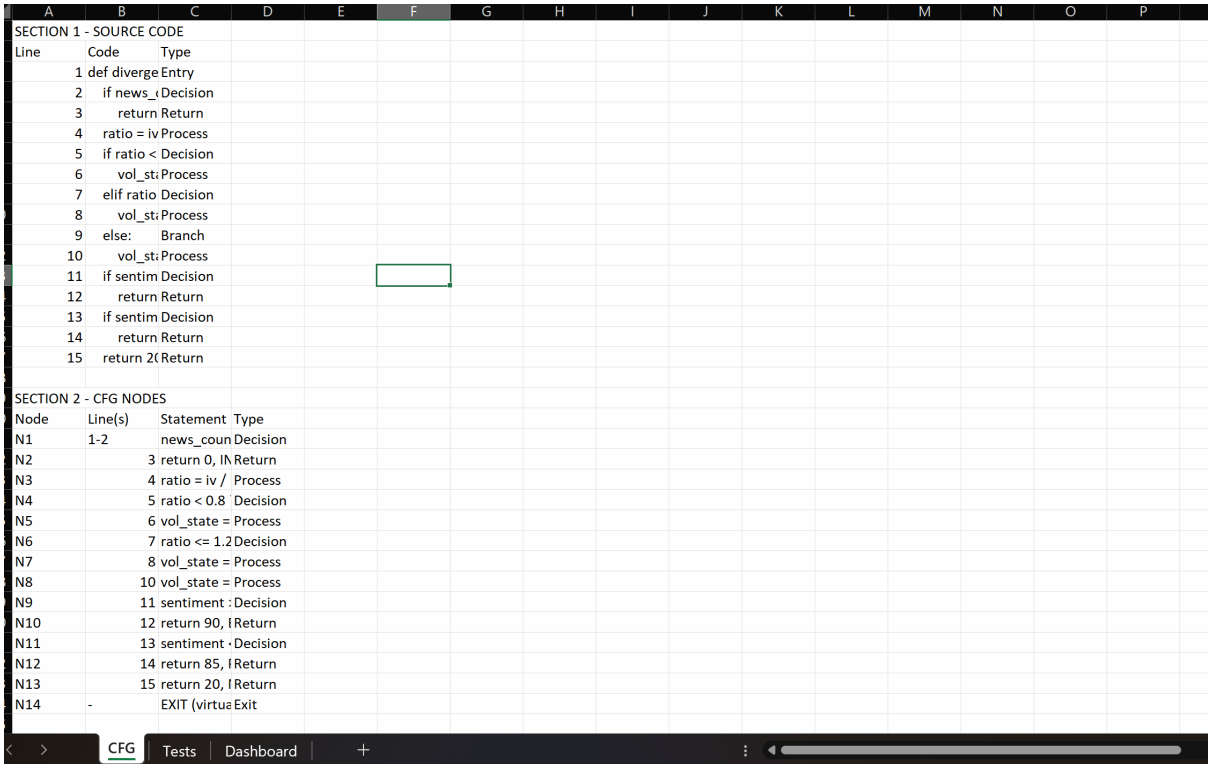
Name : Vivek ukande Roll no : 50 B4-B1



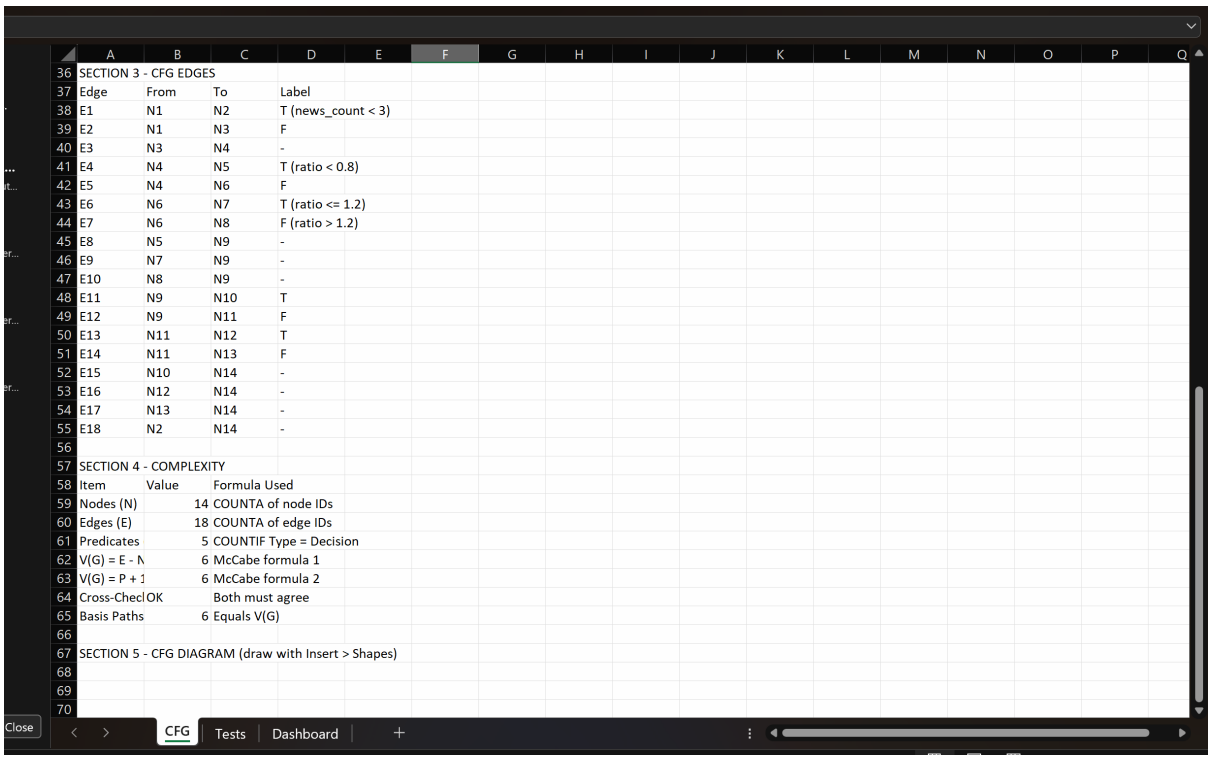
SECTION 1 - SOURCE CODE

Line	Code	Type
1	def diverge	Entry
2	if news_(	Decision
3	return	Return
4	ratio = iv	Process
5	if ratio <	Decision
6	vol_st	Process
7	elif ratio	Decision
8	vol_st	Process
9	else:	Branch
10	vol_st	Process
11	if sentim	Decision
12	return	Return
13	if sentim	Decision
14	return	Return
15	return 2(	Return

SECTION 2 - CFG NODES

Node	Line(s)	Statement	Type
N1	1-2	news_coun	Decision
N2	3	return 0, IN	Return
N3	4	ratio = iv /	Process
N4	5	ratio < 0.8	Decision
N5	6	vol_state =	Process
N6	7	ratio <= 1.2	Decision
N7	8	vol_state =	Process
N8	10	vol_state =	Process
N9	11	sentiment :	Decision
N10	12	return 90, I	Return
N11	13	sentiment ·	Decision
N12	14	return 85, I	Return
N13	15	return 20, I	Return
N14	-	EXIT (virtua	Exit

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SECTION 3 - CFG EDGES

Edge	From	To	Label
E1	N1	N2	T (news_count < 3)
E2	N1	N3	F
E3	N3	N4	-
E4	N4	N5	T (ratio < 0.8)
E5	N4	N6	F
E6	N6	N7	T (ratio <= 1.2)
E7	N6	N8	F (ratio > 1.2)
E8	N5	N9	-
E9	N7	N9	-
E10	N8	N9	-
E11	N9	N10	T
E12	N9	N11	F
E13	N11	N12	T
E14	N11	N13	F
E15	N10	N14	-
E16	N12	N14	-
E17	N13	N14	-
E18	N2	N14	-

SECTION 4 - COMPLEXITY

Item	Value	Formula Used
Nodes (N)	14	COUNTA of node IDs
Edges (E)	18	COUNTA of edge IDs
Predicates	5	COUNTIF Type = Decision
V(G) = E - N	6	McCabe formula 1
V(G) = P + 1	6	McCabe formula 2
Cross-Check	OK	Both must agree
Basis Paths	6	Equals V(G)

SECTION 5 - CFG DIAGRAM (draw with Insert > Shapes)

Close

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Name : Vivek ukande Roll no : 50 B4-B1

SECTION 1 - BASIS PATHS

Path	ID	Node Sequ	Edge Seque	Outcomes	Feasible
1	P1	N1-N2-N14	E1,E18	news_coun	Yes
2	P2	N1-N3-N4-	E2,E3,E4,E8	LOW, senti	Yes
3	P3	N1-N3-N4-	E2,E3,E4,E8	LOW, senti	Yes
4	P4	N1-N3-N4-	E2,E3,E5,E6	NORMAL, s	Yes
5	P5	N1-N3-N4-	E2,E3,E5,E7	HIGH, senti	Yes
6	P6	N1-N3-N4-	E2,E3,E5,E7	HIGH, senti	Yes

SECTION 2 - CONDITION COVERAGE (MC/DC)

ID	Predicate	Expression	TRUE by	FALSE by
C1	N9	sentiment :	TC-02 (0.8)	TC-03 (0.1)
C2	N9	vol_state =	TC-02	TC-04 (NORMAL)
N9	Decision	C1 AND C2	TC-02	TC-03, TC-04
C3	N11	sentiment ·	TC-05 (-0.9	TC-06 (-0.2)
C4	N11	vol_state =	TC-05	TC-04 (NORMAL)
N11	Decision	C3 AND C4	TC-05	TC-04, TC-06

SECTION 3 - TEST CASES

TC	Path	news_coun	sentiment	iv	hv	Expected	Actual	Status	Remarks
TC-01	P1	2	0.8	15	30	(0, INSUFFI	(0, INSUFFI	Pass	Short-circuit branch
TC-02	P2	10	0.8	15	30	(90, EUPHC	(90, EUPHC	Pass	ratio 0.50 -> LOW
TC-03	P3	10	0.1	15	30	(20, NO_DI	(20, NO_DI	Pass	Neutral sentiment, LOW
TC-04	P4	10	0.8	30	30	(20, NO_DI	(20, NO_DI	Pass	ratio 1.00 -> NORMAL
TC-05	P5	10	-0.9	45	30	(85, PANIC_	(85, PANIC_	Pass	ratio 1.50 -> HIGH
TC-06	P6	10	-0.2	45	30	(20, NO_DI	(20, NO_DI	Pass	HIGH vol, mild sentiment
TC-07	P4	10	0.8	24	30	(20, NO_DI	(20, NO_DI	Pass	ratio exactly 0.8
TC-08	P4	10	0.8	36	30	(20, NO_DI	(20, NO_DI	Pass	ratio exactly 1.2
TC-09	P5	10	0.8	36.3	30	(20, NO_DI	(20, NO_DI	Pass	ratio 1.21 -> HIGH
TC-10	P3	10	0.5	15	30	(20, NO_DI	(20, NO_DI	Pass	sentiment exactly 0.5
TC-11	P5	10	-0.5	45	30	(20, NO_DI	(20, NO_DI	Pass	sentiment exactly -0.5
TC-12	P2	3	0.8	15	30	(90, EUPHC	(90, EUPHC	Pass	news_count exactly 3
TC-13	P2	10	0.8	15	0	INVALID_V(	ZeroDivisio	Fail	Defect D-01
TC-14	P2	10	0.8	15	-30	INVALID_V(	(90, EUPHC	Fail	Defect D-02
TC-15	P3	10	NaN	15	30	DATA_QUA	(20, NO_DI	Fail	Defect D-03

SECTION 4 - COVERAGE MATRIX

Edge	TC-01	TC-02	TC-03	TC-04	TC-05	TC-06	Covered
E1	X						Yes
E2		X	X	X	X	X	Yes
E3		X	X	X	X	X	Yes
E4		X	X				Yes
E5				X	X	X	Yes
E6				X			Yes
E7					X	X	Yes
E8		X	X				Yes
E9				X			Yes
E10					X	X	Yes
E11		X					Yes
E12			X	X	X	X	Yes
E13					X		Yes
E14			X	X		X	Yes
E15		X					Yes
E16					X		Yes
E17			X	X		X	Yes
E18	X						Yes

SECTION 5 - COVERAGE SUMMARY

Metric	Value	Notes

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Name : Vivek ukande Roll no : 50 B4-B1

SECTION 4 - COVERAGE MATRIX

Edge	TC-01	TC-02	TC-03	TC-04	TC-05	TC-06	Covered
E1	X						Yes
E2		X	X	X	X	X	Yes
E3		X	X	X	X	X	Yes
E4		X	X				Yes
E5				X	X	X	Yes
E6				X			Yes
E7					X	X	Yes
E8		X	X				Yes
E9				X			Yes
E10					X	X	Yes
E11		X					Yes
E12			X	X	X	X	Yes
E13					X		Yes
E14			X	X		X	Yes
E15		X					Yes
E16					X		Yes
E17			X	X		X	Yes
E18	X						Yes

SECTION 5 - COVERAGE SUMMARY

Metric	Value	Notes
Edges Cove	18	of 18 total
Branch Cov	1	format cell as %
Statement	1	15 of 15 lines
Basis Paths	6	P1-P6 all exercised
Path Cover	#REF!	format cell as %
Condition C	1	MC/DC satisfied

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VHAGAR - WHITE BOX TESTING DASHBOARD

Prepared B Date: Version: 1.0

SECTION 1 - PROJECT INFO

Field	Value
Project	Vhagar
Module Un	divergence_score() - Mispricing Core (Feature 1)
Technique	Basis Path Testing (McCabe) + CFG + MC/DC
Language	Python 3
Workbook	3 sheets: CFG, Tests, Dashboard

SECTION 2 - SUMMARY METRICS

Metric	Value	Source
Cyclomatic	6	CFG
Basis Paths	6	CFG
Test Cases	15	Tests
Tests Passe	12	Tests
Tests Failec	3	Tests
Pass Rate %	0.8	format as %
Branch Cov	1	Tests
Path Cover	1	Tests
Defects Log	5	DefectLog
High Severi	2	DefectLog
VERDICT	HOLD - fix l	auto

SECTION 3 - DEFECT LOG

ID	Found By	Expected	Actual	Severity	Status	Remarks
D-01	TC-13	INVALID_VI	ZeroDivisio	High	Open	No hv==0 guard (line 4)
D-02	TC-14	Reject nega	ratio < 0 ->	High	Open	No input range check
D-03	TC-15	DATA_QUA	NaN compa	Medium	Open	Unsafe default on bad sentiment
D-04	TC-10 / TC-	Spec silent	Falls throu	Low	Spec Revie	Documentation gap
D-05	Inspection	news_coun	Negative cc	Low	Open	Harmless, undocumented

SECTION 4 - NOTES / METHODOLOGY

Compound predicates N9 and N11 counted at decision level (P = 5) -> V(G) = 6.

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SECTION 3 - DEFECT LOG

ID	Found By	Expected	Actual	Severity	Status	Remarks
D-01	TC-13	INVALID_VI	ZeroDivisio	High	Open	No hv==0 guard (line 4)
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D-04	TC-10 / TC-	Spec silent	Falls throu	Low	Spec Revie	Documentation gap
D-05	Inspection	news_coun	Negative cc	Low	Open	Harmless, undocumented

SECTION 4 - NOTES / METHODOLOGY

Compound predicates N9 and N11 counted at decision level (P = 5) -> V(G) = 6.

Atomic predicates would give P = 7 -> V(G) = 8; closed by MC/DC table on Tests sheet.

Basis paths P1-P6 achieve 100% statement, branch and path coverage.

Boundary TCs (07-12) probe every comparator edge; negative TCs (13-15) expose missing input validation.